

M-DaQ: Retrieving Samples with Multilingual Diversity and Quality for Instruction Fine-Tuning Datasets

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Motivation

- Multilingual IFT data is scarce & skewed toward English
- Llama-3 IFT: only 3.01% multilingual samples
- No language-agnostic quality scoring method
- SAH unverified in multilingual contexts

Challenge	M-DaQ Solution
No multilingual quality scoring	QSM (triplet loss)
Diversity limited to English	DAS (MMR-inspired)
SAH unverified multilingually	1K ~ 52K, 8 langs

Experimental Setup

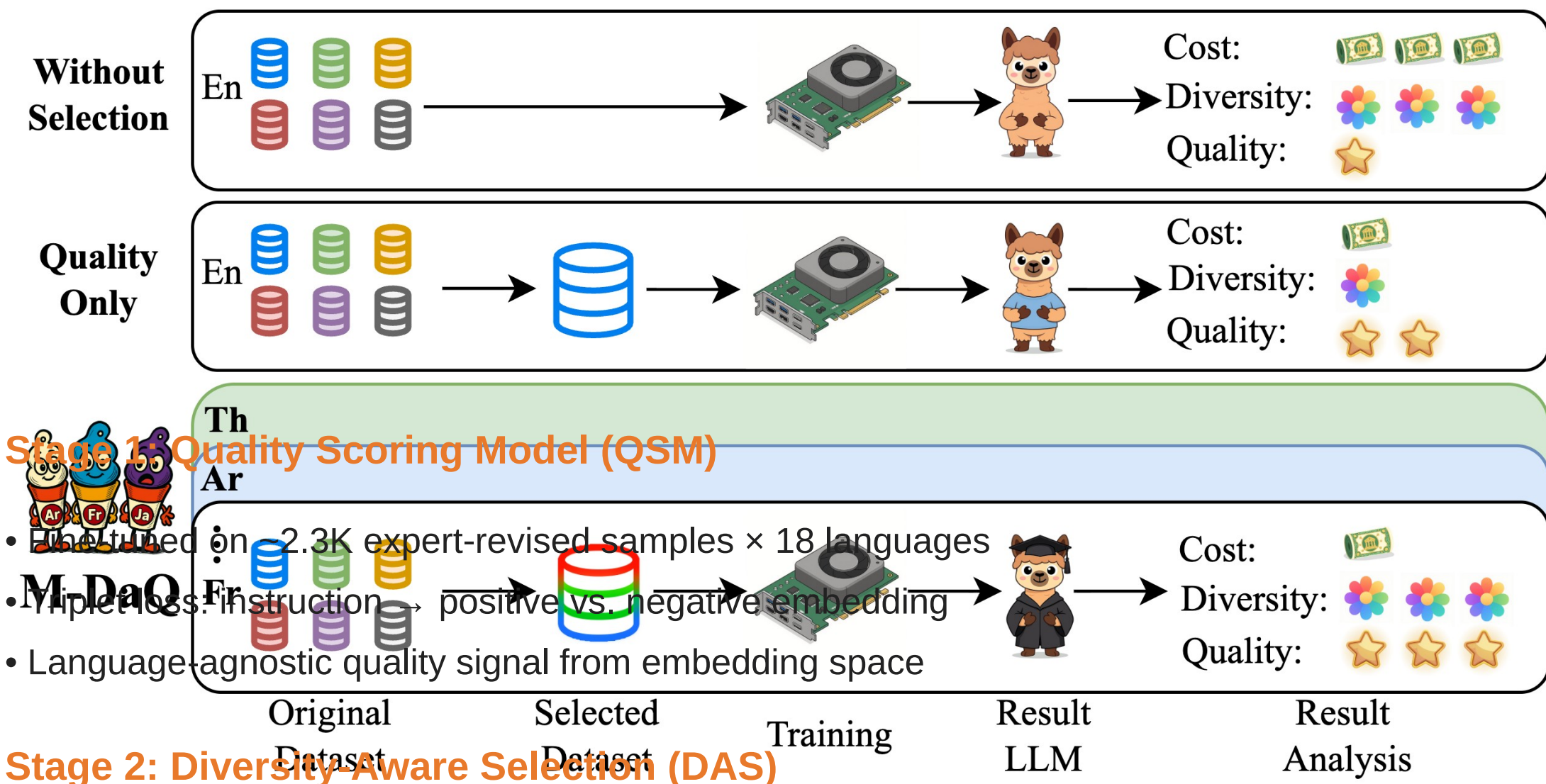
- Base model: Llama-3-8B
- IFT data: Alpaca-52K × 18 languages
- Training: 3 epochs, lr 5×10⁻⁵
- Judges: LLM-as-Judge + 7 native experts
- Benchmarks: Alpaca-Eval + MT-Bench

SAH Validation

Instruction:
Rédigez un blog de voyage captivant sur un voyage récent en Corse, en mettant en évidence les expériences culturelles et les attractions incontournables. (Write a captivating travel blog about a recent trip to Corsica, highlighting cultural experiences and must-see attractions.)

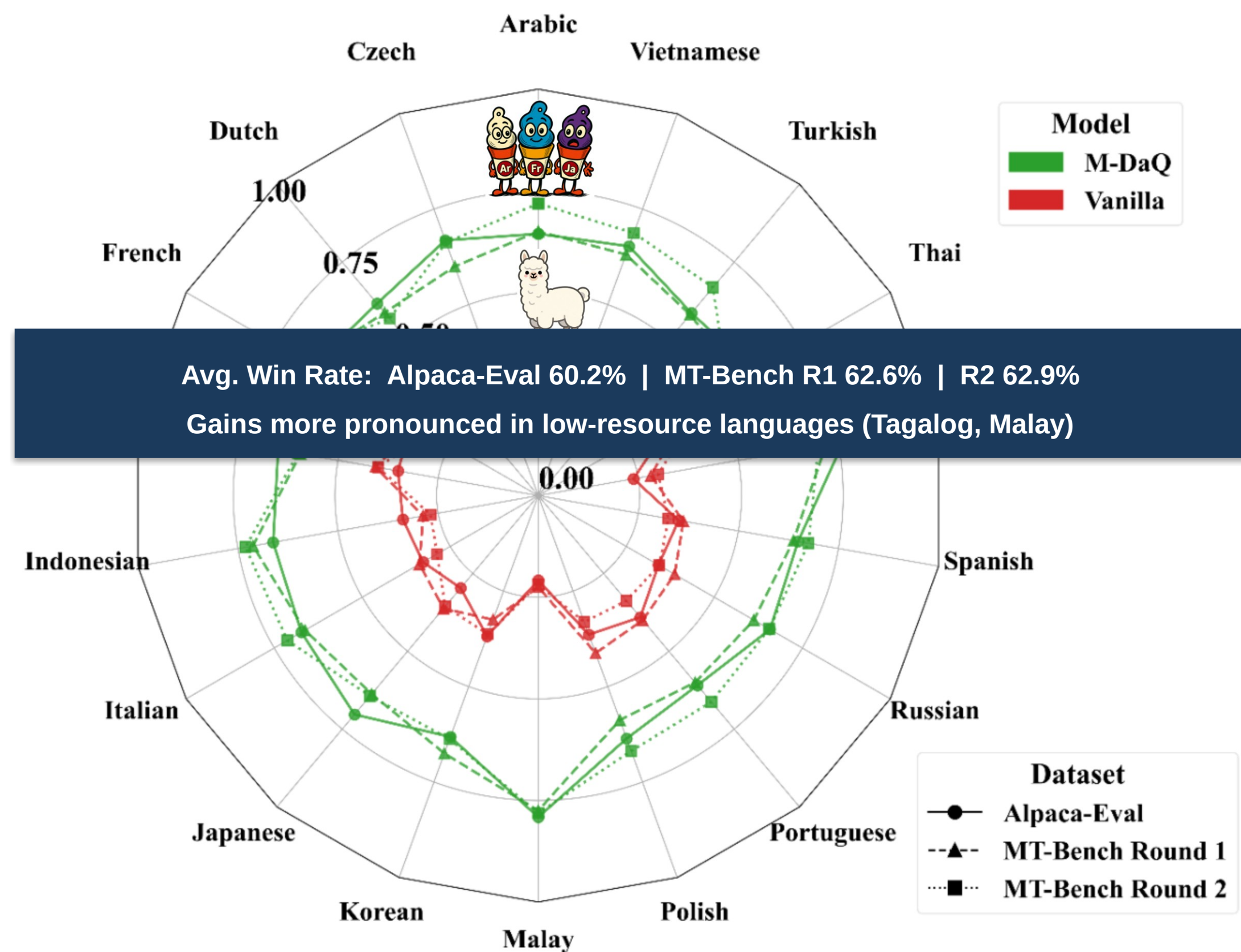
Response:
Faire un voyage en Corse est l'une des aventures ... (Making a trip to Corsica is one of the most fascinating and amazing adventures...)
L'une des expériences les plus mémorables de mon voyage a été mon visit à **Piana** (a historic mountain village in southern Corsica), un petit village perdu dans les montagnes de l'intérieur. L'histoire est fièrement animé par sa population de langue **corse** (Corsican language) et sa cuisine **traditionnelle** (Corsican traditional cuisine) reflects culture of pastoral life, Mediterranean influences). Je'ai eu la chance de goûter certains des plats locaux, compris la ravioli et la **figatelli** (a traditional Corsican dried pork sausage), et de les apprécier avec une coupe de vin rouge **corsique** (Corsican wines).

Method: M-DaQ Framework



- MMR-inspired: Quality ≈ Relevance, Diversity ≈ Novelty
- Select top-n by QSM → greedily augment from clusters
- Ratio n_quality : n_diversity = 6:1 (tuned)
- Complexity: O(n²) → O(n log n)

Cross-Lingual Results (18 Languages)



Key Results

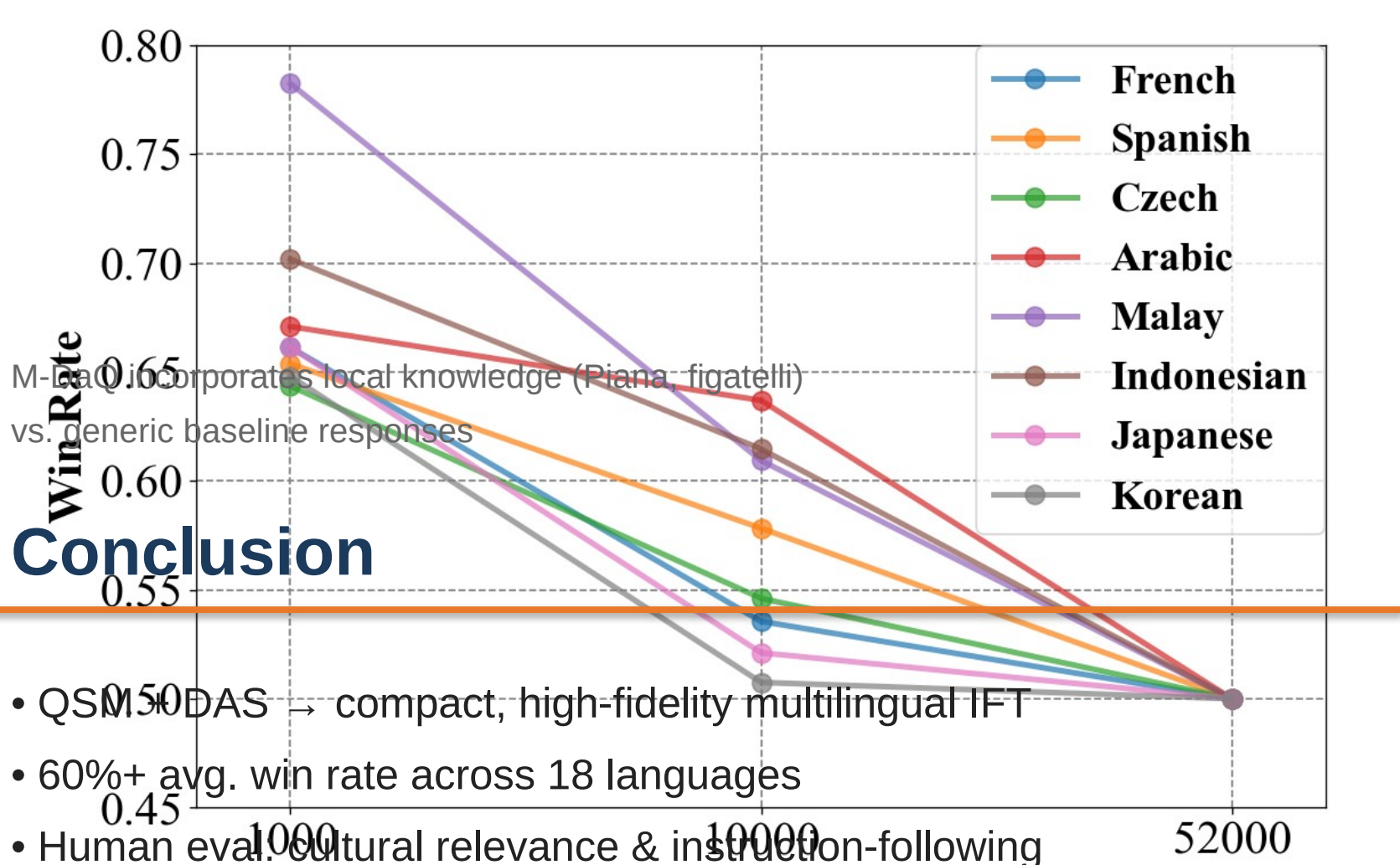
LLM-as-Judge: M-DaQ vs. Vanilla

Benchmark	Win Rate
Alpaca-Eval	60.2%
MT-Bench R1	62.6%
MT-Bench R2	62.9%

Human Eval (6 Langs, 58 Person-Hours)

Language	AE	MT-R1	MT-R2
Japanese	86%	84%	80%
Korean	94%	94%	94%
Russian	70%	86%	84%
Portuguese	80%	78%	84%
Greek	58%	76%	82%
French	80%	92%	88%
Average	78.0%	85.0%	85.3%

Cultural Localization



- QSM + DAS → compact, high-fidelity multilingual IFT
- 60%+ avg. win rate across 18 languages
- Human eval on cultural relevance & instruction-following
- First SAH validation in multilingual settings
- Few thousand quality samples > 52K unfiltered

- github.com/zhaochorey/M-DaQ
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